

■ ROCKETS & NEWTON'S 3RD LAW

A Complete Lesson Plan for Grade 2
Science · Physics · Space Technology

Subject	Science / Physics
Grade	Grade 2 (Ages 7–8)
Duration	45 – 60 minutes
Theme	Rockets · Newton's Laws · ISRO Pride
Key Concept	Action & Reaction — Newton's 3rd Law

Designed for Indian Classrooms · Aligned with NEP 2020 Principles
Inspired by ISRO's achievements — Chandrayaan, Mangalyaan & beyond!

■ LESSON OVERVIEW

Learning Objectives

- Understand Newton's 3rd Law of Motion in simple, age-appropriate language.
- Connect the law to real-life examples — jumping, swimming, rowing, and rockets.
- Develop pride in India's space achievements through ISRO's story.
- Perform a hands-on balloon rocket experiment to observe action & reaction.
- Identify and explain action–reaction pairs using their own words.

Materials Required

Item	Qty per Group	Notes
Long balloon (oblong shape)	1	Cylinder shape works best
String / nylon thread	3–5 m	Smooth, low-friction
Plastic drinking straw	1	Any standard straw
Sticky tape (Sellotape)	20 cm	For attaching balloon to straw
Chairs or poles	2	To tie the string between
Measuring tape	1 per class	Optional – to measure distance

■ BACKGROUND KNOWLEDGE

What is Newton's 3rd Law?

Sir Isaac Newton (1643–1727) was one of the greatest scientists in history. He described three fundamental rules — called **Laws of Motion** — that explain how things push, pull, and move. His **Third Law** states:

"For every action, there is an equal and opposite reaction."

In child-friendly language: **Every push has a push-back!** When you push something, it pushes YOU back — just as hard, but in the opposite direction.

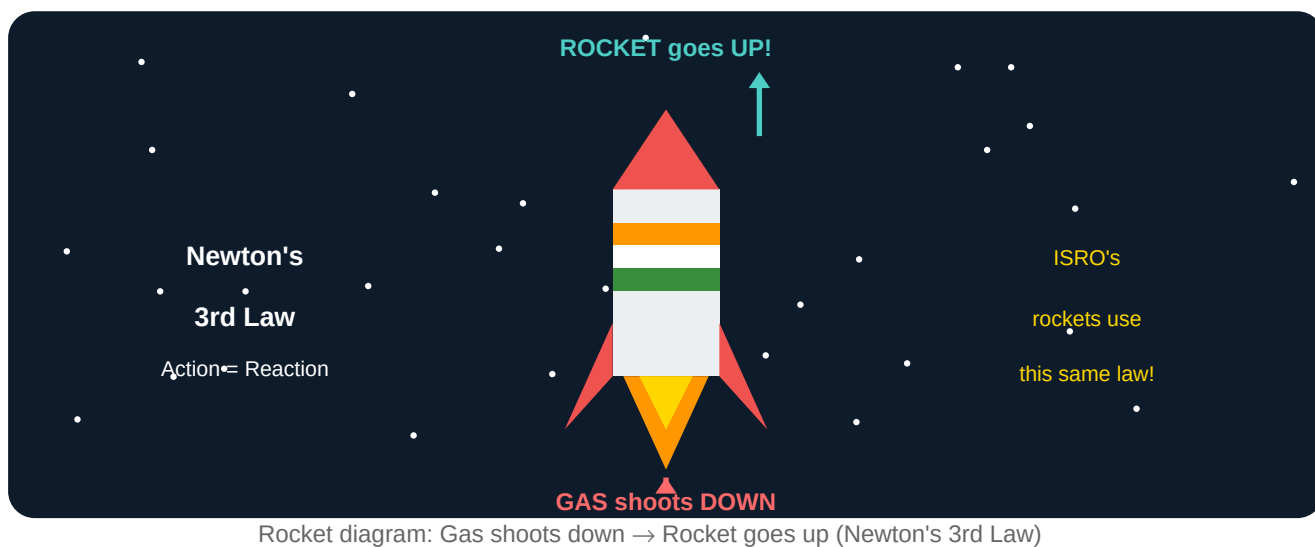
Everyday Examples of Newton's 3rd Law

Situation	Action (Push)	Reaction (Push-Back)
Jumping up	Feet push ground DOWN	Ground pushes you UP
Swimming	Arms push water BACKWARD	Water pushes body FORWARD
Rowing a boat	Oar pushes water BACKWARD	Boat moves FORWARD
Diwali rocket	Gas shoots out DOWNWARD	Rocket flies UPWARD
Clapping hands	Right hand hits left hand	Left hand hits right hand back
Walking	Foot pushes ground BACKWARD	Ground pushes foot FORWARD

How Does a Rocket Use Newton's 3rd Law?

A rocket engine burns fuel, which creates **hot gas**. This gas is pushed **downward and backward** out of the nozzle at the base of the rocket. According to Newton's 3rd Law, the gas being pushed down causes an equal and opposite push on the rocket — pushing it **upward** into the sky!

Importantly, the rocket does **NOT** need anything to push against — it carries its own fuel and oxygen. That is why rockets work even in the vacuum of space, where there is no air.



DETAILED LESSON PLAN (45–60 min)

0–5 min

Hook / Engage

Begin with the question on Slide 2: **'How does a rocket go UP if there is nothing to push against in space?'** Allow students to share guesses. Write key words on the board: PUSH, PUSH-BACK, ROCKET, SPACE.

■ **Teacher Note:** Ask: 'Have you seen a rocket launch on TV? What came out of the bottom?'

5–12 min

Explore — Push & Push-Back

Do a quick class activity. Have students clap hands once — hard. Ask: 'Did both hands feel the push?' Then invite one student to jump in place. Ask: 'You pushed the ground DOWN — what happened to you?'

Introduce the phrases **Action** and **Reaction**. Write: **Action** → **Reaction** on board.

■ **Teacher Note:** Use slides 3–4. Ask students to point ACTION hand and REACTION hand.

12–20 min

Explain — Newton's 3rd Law

Formally introduce Newton's 3rd Law (Slide 4). Explain in simple language: *'When you push something, it pushes you back — equally hard, in the OPPOSITE direction!'*

Go through Indian examples on Slide 5: rowing boat, Diwali rockets, high-five clap. For each, ask students to identify the Action and the Reaction.

■ **Teacher Note:** Reinforce: 'PUSH = PUSH BACK!' Repeat it together as a class chant.

20–32 min

Experiment — Balloon Rocket

Set up the balloon rocket experiment (Slides 6–8). Divide class into groups of 4–5. Each group: threads string through straw, ties string between two chairs, blows balloon (don't tie), tapes balloon to straw, and releases.

Observe: Air rushes backward → Balloon shoots forward. **This IS Newton's 3rd Law!**

■ **Teacher Note:** Ask groups to discuss: 'What was the Action? What was the Reaction?'

32–40 min

Connect to Real Rockets

Use Slide 10 to connect the experiment to a real rocket. **Balloon experiment:** Air goes back → Balloon goes forward.

Real rocket: Gas goes down → Rocket goes up!

Introduce rocket parts: Nose Cone, Body, Nozzle. Draw on the board.

■ **Teacher Note:** Show rocket diagram. Ask: 'What comes out of the nozzle?'

40–47 min

ISRO Pride

Celebrate India's achievements (Slide 11): PSLV launched 104 satellites in ONE launch (world record!); Chandrayaan-3 landed on the Moon's south pole (2023); Mangalyaan reached Mars on the FIRST try (2014); Dr. A.P.J. Abdul Kalam — the Missile Man of India. Tell students: '**All of these used Newton's 3rd Law!**'

■ **Teacher Note:** Inspire: 'You can be a future ISRO scientist!'

47–55 min Review — Rocket Riddles

- Use Slides 12–13 for quick oral quiz (Rocket Riddles):
- Rocket pushes gas DOWN → Which way does rocket go? (UP!)
 - Boatman pushes water BACK → Boat goes? (FORWARD!)
 - Name this law? (Newton's 3rd Law)
 - Which Indian space agency uses this? (ISRO!)

■ **Teacher Note:** Raise-your-hand format. Celebrate correct answers!

55–60 min Wrap-Up & Assessment

Close with Slide 14: '**JAI HIND! JAI VIGYAN!**'
Ask each student to verbally complete: '*When I push _____, it pushes me _____!*'
Distribute worksheet (see last page) for individual practice.

■ **Teacher Note:** Optional: Assign a draw-and-label homework of the rocket diagram.

BALLOON ROCKET EXPERIMENT — STEP BY STEP

Step 1 Thread the String	Take the plastic straw and thread the full length of string through it (from one end to the other). The straw should slide freely along the string.
Step 2 Set Up the Track	Tie one end of the string to the back of a chair. Stretch the string tightly across the room and tie the other end to another chair. Make sure the string is taut and horizontal .
Step 3 Inflate the Balloon	Blow up a long balloon as much as possible. Do NOT tie the opening . Pinch the opening shut with your fingers to keep the air inside.
Step 4 Attach Balloon to Straw	While still holding the balloon opening closed, tape the inflated balloon to the straw (which is already on the string). Use two pieces of tape — one near each end of the straw.
Step 5 Position the Balloon	Slide the straw (and balloon) to one end of the string. Make sure the closed end of the balloon faces the direction you want it to travel.
Step 6 BLAST OFF!	Count down together: 3... 2... 1... BLAST OFF! Let go of the balloon opening. Watch the air rush out the back and the balloon zoom forward along the string!



Balloon Rocket Experiment — Newton's 3rd Law in Action!

Diagram: Air goes back (Action) → Balloon goes forward (Reaction)

What's Happening? (The Science)

When you release the balloon, pressurised air rushes **OUT of the back** (this is the **Action**). By Newton's 3rd Law, this Action creates an equal and opposite **Reaction** — which pushes the balloon **FORWARD**. This is **exactly** how a rocket engine works: gas shoots out the nozzle downward (Action) → rocket flies upward (Reaction)!

■■ ISRO PRIDE — INDIA'S SPACE CHAMPIONS

About ISRO

The **Indian Space Research Organisation (ISRO)**, founded in 1969, is India's national space agency headquartered in Bengaluru. It is one of the top six space agencies in the world, known for conducting space missions at a fraction of the cost of other agencies. Every ISRO rocket uses Newton's 3rd Law!

Mission / Achievement	Year	Why It's Amazing	Newton's 3rd Law
PSLV — 104 Satellites in One Launch	2017	World record! Launched 104 satellites in a single rocket.	Rocket exhaust pushed DOWN → Rocket soared UP!
Chandrayaan-3 (Moon Landing)	2023	First to land on Moon's south pole — ever!	Descent engines fired DOWN → Lander slowed & settled.
Mangalyaan (Mars Orbiter)	2014	Reached Mars on the first attempt — and cheapest!	Rocket thrust carried payload 150 million km to Mars.
Chandrayaan-1 (Moon Orbiter)	2008	Discovered water molecules on the Moon!	Multi-stage rocket used 3rd Law at every stage.
AstroSat (Space Telescope)	2015	India's first dedicated astronomy satellite.	PSLV rocket (Newton's 3rd Law) put it in orbit.

Dr. A.P.J. Abdul Kalam — The Missile Man of India

Dr. Avul Pakir Jainulabdeen Abdul Kalam (1931–2015) was a legendary aerospace scientist and the 11th President of India. He led India's first Satellite Launch Vehicle (SLV-III) programme and was the chief architect of India's missile development programme (Project Devil, Project Valiant, and Agni). He inspired millions of children to **dream big** and pursue science. His most famous quote: *'Dream, Dream, Dream. Dreams transform into thoughts and thoughts result in action.'* Every rocket he designed was powered by Newton's 3rd Law!

⇒ ■ STUDENT WORKSHEET

Name: _____ Date: _____ Class: _____

Part A — Fill in the Blanks

- Newton's 3rd Law says: For every _____, there is an equal and opposite _____.
- When you jump, you push the ground _____ and the ground pushes you _____.
- When a rocket pushes gas _____, the rocket moves _____.
- The letters ISRO stand for: Indian _____ Research Organisation.
- In our balloon experiment, the air went _____ so the balloon went _____.

Part B — Match the Action to the Reaction

ACTION		REACTION
Rocket pushes gas DOWN	→	_____
You push ground DOWN	→	_____
Oar pushes water BACKWARD	→	_____
You push a wall	→	_____

Part C — Think & Write

Q1. In your own words, what is Newton's 3rd Law of Motion? (2 sentences)

Q2. How is our balloon rocket experiment similar to a real ISRO rocket launch?

Q3. Name ONE achievement of ISRO that you found most exciting. Why?

Part D — Draw & Label the Rocket

Draw a rocket below. Label: **Nose Cone**, **Body**, **Nozzle**, **Gas going DOWN**, and **Rocket going UP**. Draw arrows to show the Action and Reaction.



ANSWER KEY (Teacher's Copy)

Part A — Fill in the Blanks (Answers)

- ✓ 1. action | reaction
- ✓ 2. DOWN | UP
- ✓ 3. DOWN / backward | UP / forward
- ✓ 4. Space
- ✓ 5. backward | forward

Part B — Match the Action to the Reaction (Answers)

- ✓ Rocket pushes gas DOWN → Rocket goes UP
- ✓ You push ground DOWN → You go UP
- ✓ Oar pushes water BACKWARD → Boat goes FORWARD
- ✓ You push a wall → Wall pushes you BACK

Part C — Suggested Answers

Q1. Newton's 3rd Law says that every action has an equal and opposite reaction. When you push something, it pushes back just as hard in the opposite direction.

Q2. In both cases, something is pushed backward (air / gas) which causes the object (balloon / rocket) to move forward. Both demonstrate Action and Reaction.

Q3. Accept any valid answer. Encourage students to explain why (curiosity, national pride, scientific importance). Example: 'Chandrayaan-3 — because no one had ever landed on the Moon's south pole before!'

Assessment Rubric

Criteria	4 — Excellent	3 — Good	2 — Developing	1 — Needs Support
Understands Law	Can explain law clearly in own words with examples	Explains law with some examples	Partially explains the law	Cannot explain the law
Experiment Participation	Actively leads, explains observations	Participates fully	Participates with prompting	Needs significant support
Connecting to ISRO	Links law to ISRO missions in detail	Mentions ISRO and one mission	Mentions ISRO	Cannot connect to ISRO
Worksheet Completion	All correct, clear and neat	Mostly correct	Some correct with errors	Mostly incomplete

Extension / Homework Activities

- **Art & Science:** Draw and colour a labelled diagram of an ISRO rocket, showing the nozzle, gas exhaust, and direction arrows for Action and Reaction.

- **Creative Writing:** Write a short story: 'I am an ISRO scientist. Today we are launching Chandrayaan-4...'
- **Experiment Variation:** Try the balloon rocket with different balloon sizes. Which goes further? Why?
- **Video Research:** Watch a video of an ISRO rocket launch and identify the Action and Reaction.
- **Math Connection:** Measure the distance the balloon travels in the experiment. Record results in a table.

■ JAI HIND! JAI VIGYAN! — Dream Big, You Are a Future ISRO Scientist! ■